DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: CS301 - Computer Organization & Architecture

Lab Assignment: Cache Coherence Protocols in Multicore Processors

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1. Abstract & Problem Statement

In shared-memory multiprocessor systems, individual processors maintain local private caches to reduce memory latency. This introduces the cache coherence problem when multiple caches hold copies of the same memory address. This paper evaluates the MESI (Modified, Exclusive, Shared, Invalid) write-invalidate protocol under high contention.

2. MESI State Transition Diagram & Simulation Results

We simulated 8-core CPU executing parallel SPLASH-2 benchmarks. Snooping latency across pipelined bus was measured. Write hits in Modified state incur 0 cycle bus overhead, while Shared-to-Modified transitions require invalidation broadcast. Average cache line invalidations: 4,120 per million clock cycles. Speedup attained: 6.4x on 8 cores.

3. Conclusion & Student Reflections

Directory-based coherence protocols scale significantly better than snooping buses for core counts exceeding 16. Submitted as partial fulfillment of laboratory assessment curriculum requirements.

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